ANALYTICAL MODELING & PI CONTROLLER DESIGN FOR MAXON RE 25 DC MOTOR

Component Part No.: 339152 **Application:** Speed Control with Load Disturbance Rejection

1. EXECUTIVE SUMMARY & SYSTEM OVERVIEW

This technical report details the complete step-by-step physical modeling, transfer function derivation, and analytical PI controller design for the **Maxon RE 25 DC Motor (Part Number: 339152)**. To ensure exact matching with physical hardware behavior, the full second-order dynamic model including armature inductance (L_a) is maintained. A robust PI controller is derived analytically using the *Pole Placement Method*, guaranteeing specified bandwidth, zero steady-state error, and disturbance rejection under full nominal load torque ($T_L = 27.1\ mNm$).

2. MOTOR PHYSICAL SPECIFICATIONS & SI CONVERSIONS

The parameters are extracted directly from the official Maxon RE 25 catalog datasheet (339152) and converted into standard International System of Units (SI):

Parameter Name	Symbol	Datasheet Value	Standard SI Value
Nominal Operating Voltage	V_n	24.0 V	**24.0 V**
Armature Resistance	R_a	1.53 Ω	**1.53 Ω**
Armature Inductance	L_a	0.186 mH	**0.000186 H** (1.86×10^{-4} H)
Torque Constant	K_t	20.8 mNm/A	**0.0208 N·m/A**
Back-EMF Constant	K_e	459 rpm/V	**0.0208 V·s/rad**
Rotor Inertia	J	14.7 g·cm²	**1.47×10^{-6} kg·m²**
Mechanical Time Constant	τ_m	5.24 ms	**0.00524 s**
Nominal Torque	$T_{Nominal}$	27.1 mNm	**0.0271 N·m**
Nominal Speed	$\omega_{Nominal}$	9690 rpm	**1014.72 rad/s**
No-load Speed	$\omega_{No\text{-}load}$	10900 rpm	**1141.45 rad/s**

3. DYNAMIC GOVERNING DIFFERENTIAL EQUATIONS

The electromechanical behavior of the DC motor is governed by Kirchhoff's Voltage Law (KVL) for the electrical circuit and Newton's Second Law for rotational mechanics:

ELECTRICAL CIRCUIT EQUATION:

$$V(t) = L_a \frac{d\, i_a(t)}{dt} + R_a\, i_a(t) + e_b(t) \quad \text{where} \quad e_b(t) = K_e\, \omega(t)$$

MECHANICAL MOTION EQUATION:

$$J \frac{d\, \omega(t)}{dt} + b\, \omega(t) = T_m(t) - T_L(t) \quad \text{where} \quad T_m(t) = K_t\, i_a(t)$$

Note: Viscous damping friction b is negligible ($b \approx 0$) relative to the magnetic back-EMF damping coefficient.

4. DERIVATION OF FULL 2ND-ORDER TRANSFER FUNCTIONS

Applying the Laplace Transform under zero initial conditions ($i_a(0) = 0$, $\omega(0) = 0$):

$$I_a(s) = \frac{V(s) - K_e\,\Omega(s)}{L_a\,s + R_a} \qquad \text{and} \qquad J\,s\,\Omega(s) = K_t\,I_a(s) - T_L(s)$$

Substituting $I_a(s)$ into the mechanical dynamics equation yields the multi-input system relation:

$$\Omega(s) = \frac{K_t}{L_a\,J\,s^2 + R_a\,J\,s + K_e\,K_t}\,V(s) + \frac{-(L_a\,s + R_a)}{L_a\,J\,s^2 + R_a\,J\,s + K_e\,K_t}\,T_L(s)$$

By Superposition ($\Omega(s) = G_v(s)\,V(s) + G_L(s)\,T_L(s)$), the two individual transfer functions are defined:

VOLTAGE-TO-SPEED TRANSFER FUNCTION:

$$G_v(s) = \frac{\Omega(s)}{V(s)} = \frac{K_t}{L_a\,J\,s^2 + R_a\,J\,s + K_e\,K_t}$$

LOAD TORQUE TRANSFER FUNCTION:

$$G_L(s) = \frac{\Omega(s)}{T_L(s)} = \frac{-(L_a\,s + R_a)}{L_a\,J\,s^2 + R_a\,J\,s + K_e\,K_t}$$

5. NUMERICAL CALCULATION OF MOTOR COEFFICIENTS

Substituting the SI values of Maxon RE 25 (339152) into the coefficients:

- **Coefficient of s^2:** $L_a\,J = (0.000186) \times (1.47 \times 10^{-6}) =$ **2.7342×10^{-10} kg·m²·H**
- **Coefficient of s:** $R_a\,J = (1.53) \times (1.47 \times 10^{-6}) =$ **2.2491×10^{-6} kg·m²·Ω**
- **Constant term (s^0):** $K_e\,K_t = (0.0208) \times (0.0208) =$ **0.00043264 N·m·V·s/rad·A**
- **Voltage Numerator (K_t): 0.0208**

Final Numerical Plant Transfer Functions:

$$G_v(s) = \frac{0.0208}{2.7342 \times 10^{-10}\,s^2 + 2.2491 \times 10^{-6}\,s + 0.00043264}$$

$$G_L(s) = \frac{-0.000186\,s - 1.53}{2.7342 \times 10^{-10}\,s^2 + 2.2491 \times 10^{-6}\,s + 0.00043264}$$

6. ANALYTICAL PI CONTROLLER DESIGN (POLE PLACEMENT METHOD)

The Proportional-Integral (PI) controller is defined as:

$$C(s) = K_p + \frac{K_i}{s} = \frac{K_p\, s + K_i}{s}$$

Normalizing $G_v(s)$ to standard monic polynomial form $G_v(s) = \frac{b_0}{s^2 + a_1 s + a_0}$:

$$b_o = \frac{K_t}{L_a J} = \frac{0.0208}{2.7342 \times 10^{-10}} \approx \mathbf{7.6073 \times 10^{7}}$$

$$a_1 = \frac{R_a}{L_a} = \frac{1.53}{0.000186} \approx \mathbf{8225.81}$$

$$a_o = \frac{K_e\, K_t}{L_a J} = \frac{0.00043264}{2.7342 \times 10^{-10}} \approx \mathbf{1.5823 \times 10^{6}}$$

The overall Closed-Loop Characteristic Equation ($1 + C(s)\, G_v(s) = 0$) forms a 3rd-order polynomial:

$$s^3 + a_1\, s^2 + (a_o + b_o\, K_p)\, s + b_o\, K_i = 0$$

We equate this characteristic polynomial with a target 3rd-order polynomial comprising a dominant complex conjugate pole pair (ζ, ω_n) and a real pole (p):

$$(s + p)(s^2 + 2\zeta\, \omega_n\, s + \omega_n{}^2) = s^3 + (2\zeta\, \omega_n + p)\, s^2 + (\omega_n{}^2 + 2\zeta\, \omega_n\, p)\, s + p\, \omega_n{}^2 = 0$$

Equating Coefficients:

$$1)\quad 2\zeta\, \omega_n + p = a_1 \;\Rightarrow\; \boldsymbol{p = a_1 - 2\zeta\, \omega_n}$$

$$2)\quad \omega_n{}^2 + 2\zeta\, \omega_n\, p = a_o + b_o\, K_p \;\Rightarrow\; \boldsymbol{K_p = \frac{\omega_n{}^2 + 2\zeta\, \omega_n\, p - a_o}{b_o}}$$

$$3)\quad p\, \omega_n{}^2 = b_o\, K_i \;\Rightarrow\; \boldsymbol{K_i = \frac{p\, \omega_n{}^2}{b_o}}$$

7. EXACT NUMERICAL COMPUTATION OF K_P AND K_I

Design Specifications: Damping ratio $\zeta = 0.707$ (optimal response with minimal overshoot) and natural frequency $\omega_n = 500$ *rad/s* (expanding closed-loop bandwidth ~2.5x beyond uncompensated dynamics).

STEP 1: REAL POLE LOCATION (P)

$$p = 8225.81 - 2(0.707)(500) = 8225.81 - 707.00 = \mathbf{7518.81\ rad/s}$$

STEP 2: PROPORTIONAL GAIN (K_P)

$$\omega_n{}^2 = 500^2 = 250{,}000$$

$$2\zeta\,\omega_n\,p = 707 \times 7518.81 = 5{,}315{,}798.67$$

$$K_p = \frac{250{,}000 + 5{,}315{,}798.67 - 1{,}582{,}300}{7.6073 \times 10^7} = \frac{3{,}983{,}498.67}{7.6073 \times 10^7} \approx \mathbf{0.05236}$$

STEP 3: INTEGRAL GAIN (K_I)

$$K_i = \frac{7518.81 \times 250{,}000}{7.6073 \times 10^7} = \frac{1{,}879{,}702{,}500}{7.6073 \times 10^7} \approx \mathbf{24.709}$$

PI Controller Gain	Analytical Calculation	Simulink Block Value	Functional Role
Proportional Gain (K_p)	**0.05236**	`0.0524`	Sets rise time and bandwidth
Integral Gain (K_i)	**24.709**	`24.71`	Eliminates steady-state error & rejects disturbance

8. MATLAB/SIMULINK IMPLEMENTATION & DISTURBANCE REJECTION

The system is constructed using standard Simulink blocks using the Superposition Method:

- **Reference Speed Signal:** `Step` block with initial value = 0, final value = `1014.5` rad/s (≈ 9690 rpm), step time = 0.0 s.
- **PI Controller Block:** `PID Controller` block configured in PI mode with *P = 0.0524* and *I = 24.71*.
- **Voltage Saturation:** `Saturation` block placed after PI controller with limits set to **+24V** / **-24V**. Anti-windup method enabled via **Clamping** or **Back-calculation**.
- **Plant Blocks:** Two `Transfer Fcn` blocks representing $G_v(s)$ and $G_L(s)$.
- **Load Disturbance Input (T_L):** `Step` block with Step Time = `3.0` s, Initial Value = `0`, Final Value = `0.0271` N·m.

Expected Closed-Loop Performance:
Upon execution, the motor accelerates from rest to 1014.5 rad/s within $t < 15$ *ms* with < 5% overshoot. When full nominal load torque (27.1 mNm) is abruptly applied at $t = 3.0$ *s*, an instantaneous minor dip in speed occurs, which the integral action ($K_i = 24.71$) restores back to exactly 1014.5 rad/s in less than 50 ms.